Exercise 2 – Exploring LLMs

1. Current Rank of My Favourite LLMs

First, my favourite LLM is Claude Opus 5(max). It was developed by Anthropic, because it is strong at reasoning, coding, and detailed technical explanations.

As of 24 August 2026, it is ranked 1st out of 169 models on the Artificial Analysis Intelligence Leadboard, with an Intelligence index score of 63

And I also checked GPT-5.6 Sol (max) by OpenAI; this is ranked 5th on the same Leadboard, with an Intelligence index score of 61

2. Two New Models Selected

I selected two LLMs that I had not used before:

1.Qwen3.8 Max

It was developed by Alibaba

Website: chat.qwen.ai

2.DeepSeek V4 Flash 0731(max)

It was developed by DeepSeek

Website:chat.deepseek.com

Prompt 1

Deepseek

Since you asked again, I'll give you a **fresh, super-concise take** with a different analogy and new examples, so you get a well-rounded view.

The Core Difference in One Sentence:

- **Supervised** = You give the computer **both the questions AND the answers** upfront, so it learns to map one to the other.

- **Unsupervised** = You give the computer **only the questions** and let it find interesting patterns all by itself.

The Driving Instructor Analogy:

- **Supervised** is like learning to drive with an instructor who constantly tells you, *"Turn left here,"* *"Brake now,"* *"That's a stop sign."* You learn the rules by matching your actions to their labels. Later, you drive alone and make the right moves.

- **Unsupervised** is like being dropped into a new city with no map. You drive around and notice, *"Hey, most of the coffee shops are on the east side,"* and *"The roads get busier around 5 PM."* No one told you that—you just spotted the clusters yourself.

Two New Real-World Examples:

Type	Real-World Example
Supervised	**House Price Prediction** – You feed the computer a spreadsheet of 10,000 houses. Each row has features (size, bedrooms, location) AND the actual sale price (the "label"). The computer learns the relationship. When a new house hits the market, it predicts its price.

| **Unsupervised** | **News Article Grouping** – A news website feeds thousands of today's articles into a computer without any categories. The computer reads the text and automatically groups them into clusters: sports stories go together, political stories go together, and tech stories go together—all without anyone telling it what "sports" or "politics" means. |

The Ultimate Cheat Sheet:

- **Supervised** needs **labeled data** (costly, takes human effort).

- **Unsupervised** uses **unlabeled data** (cheap, plentiful).

- **Supervised** answers: *"What will happen next?"* (prediction).

- **Unsupervised** answers: *"What is hidden in here?"* (discovery).data; unsupervised learning finds hidden patterns in unlabelled data.

Explain the difference between supervised learning and unsupervised learning

Thought for 4 seconds

1. The user is asking the same question as before: "Explain the difference between supervised learning and unsupervised learning."
2. The previous assistant response gave a detailed, clear explanation with examples.
3. The user might be asking again because they want a different angle, a shorter summary, a more technical definition, or they didn't read the previous response (maybe it was cut off? But it wasn't). Given that the prompt is identical, I should provide a new, concise, and fresh explanation to avoid repetition. I can add a bit more technical nuance (e.g., regression vs. classification, clustering vs. association) or use different analogies and examples to add value.
4. Let's craft a response that is slightly different but equally clear. I can use the "labeled vs. unlabeled" data core distinction. I can give examples like house price prediction (supervised) and news topic clustering (unsupervised). I can also add a quick comparison table or a one-sentence summary.
5. I will keep it simple, but ensure it's not a verbatim copy of the previous response. Let's provide a fresh take.

Since you asked again, I'll give you a **fresh, super-concise take** with a different analogy and new examples, so you get a well-rounded view.

Qwen3.8-Max

Deep Research...

Research Planning

This research aims to deliver a thorough analysis of the differences between supervised and unsupervised learning across multiple dimensions. It includes both conceptual explanations and real-world practical examples, with applications spanning domains like image recognition and customer segmentation. The comparison should be accessible yet detailed enough to serve diverse audiences—beginners seeking foundational understanding, technical practitioners needing implementation insights, and business decision-makers evaluating use cases. Key distinguishing factors to be addressed include the nature of input data (labeled vs. unlabeled), types of outputs (predictions vs. patterns/clusters), commonly used algorithms (e.g., regression, neural networks vs. k-means, PCA), and methods for evaluating model performance. The scope encompasses all major aspects implicitly requested by the user's 'all' response, ensuring no dimension is omitted while maintaining clarity and relevance across contexts. No temporal constraints are specified, so the analysis remains current and timeless unless otherwise implied by standard practices in machine learning education and application.

Uncovering Core Conceptual Divide: Labeled Guidance vs. Unlabeled Structure Discovery in ML

📘 **Foundational Distinction Clarified**: Supervised learning fundamentally relies on *ground-truth labels*—explicit target outputs (e.g., class tags, regression values) that supervise the model's function approximation task. Unsupervised learning operates exclusively on *unlabeled data*, where no predefined outputs exist, and the objective shifts to discovering intrinsic structure—such as clusters, latent patterns, or dimensional reductions—without external guidance.

🎯 **Learning Objective Contrast**:

- Supervised: Learns a mapping *f: X → Y* where Y is known during training (e.g., predicting tumor malignancy from biopsy features).
- Unsupervised: Discovers hidden relationships *within X* alone—like grouping customers by purchasing behavior (clustering) or compressing image representations (PCA, autoencoders).

🔍 **Epistemic Boundary Defined**: The presence—or absence—of human-annotated labels constitutes the decisive epistemological boundary: supervised methods assume verifiable truth exists for each input; unsupervised methods assume truth is emergent, relational, and must be inferred from data geometry and statistical regularity.

📚 **Source Consensus**: Cross-validated across authoritative ML textbooks (e.g., Bishop's *Pattern Recognition and Machine Learning*), Stanford CS229 lecture notes, and scikit-learn's official conceptual documentation—this distinction is consistently framed not as a technical implementation difference, but as a *paradigm-level divergence in learning philosophy and data dependency*.

Prompt 2

Write a simple Python program that accepts five numbers from the user and displays their average. Explain the code briefly.

DeepSeek -response

The Code

python

```python
# Initialize a variable to store the total sum
total = 0

# Ask the user for 5 numbers using a loop
for i in range(1, 6):
    # Prompt the user for each number
    num = float(input(f"Enter number {i}: "))
    # Add the number to the running total
    total += num

# Calculate the average by dividing total by 5
average = total / 5

# Display the result
print(f"\nThe average of the five numbers is: {average}")
```

Brief Explanation

Line(s)	What it does
`total = 0`	Creates a variable to keep a running sum of all entered numbers.
`for i in range(1, 6):`	Loops 5 times (i goes from 1 to 5). This lets us ask for each number one by one.
`float(input(f"Enter number {i}: "))`	Takes user input, converts it to a decimal number (so that `3` and `3.5` both work), and stores it in `num`.
`total += num`	Adds the current number to the running total (same as `total = total + num`).
`average = total / 5`	After the loop finishes, divides the sum by 5 to get the average.
`print(...)`	Displays the final average in a clear message.

Example Run

Enter number 1: 10

Enter number 2: 20

Enter number 3: 30

Enter number 4: 40

Enter number 5: 50

The average of the five numbers is: 30.0

A small improvement: If you want to handle the case where the user types text instead of a number, you could add a `try/except` block—but for a simple program, this works perfectly!

Qwen3.8-Max response

```python

# Program to find the average of five numbers
```

```
total = 0

for i in range(5):
    num = float(input("Enter number " + str(i + 1) + ": "))
    total += num

average = total / 5

print("The average is:", average)
```

Explanation

- `total = 0` creates a variable to store the sum of the five numbers.

- The `for` loop runs five times to get five inputs from the user.

- `float(input(...))` converts the user input into a decimal number.

- Each number is added to `total`.

- After the loop, the average is calculated by dividing `total` by `5`.

- Finally, the result is displayed using `print()`.

Qwen3.8-Max

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